# Augmented Block Designs
# How to (roughly) screen large numbers of accessions

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# Augmented Block Designs

- Introduced by Federer (1956)
- Controls (check varieties) are replicated in a standard experimental design : Mostly block designs, but also split-plot designs
- New treatments (genotypes) are not replicated, or have fewer replicates than the checks – they **augment** the standard design
- Provide an estimate of standard error that can be used for comparisons
  - Among the new genotypes
  - Between new genotypes and check varieties
- Observations on new genotypes can be adjusted for field heterogeneity (blocking)
- **Unreplicated designs can make good use of scarce resources**
- Flexible – blocks can be of unequal size

# Setup of an augmented design, based on a RCBD

- Area is divided into blocks
  these are incomplete blocks because they contain only a subset of the entries
- Two or more check varieties are assigned at random to plots within the blocks
  same check varieties appear in each block
  little is lost if you want to place one check systematically - a block marker
- Most efficient when block size is constant
- Checks are replicated, but new entries are not

# Drawbacks

- Considerable resources are spent on production and processing of control plots
- Relatively few degrees of freedom for experimental error, which reduces the power to detect differences among treatments
- Unreplicated experiments are inherently imprecise, no matter how sophisticated the design

# Applications

- Early stages in a breeding program
- Evaluation of large biodiversity collections
  - May be insufficient seed for replication
  - Using a single replication permits more genotypes to be screened
- Participatory plant breeding or biodiversity evaluation (common gardens)
- Want to evaluate promising genotypes (or other technologies) in as many environments as possible

# Defining the number of blocks

- at least 10 ddl for residual variance of the underlying RCBD
- # of ddl of residual variance : $(r-1)(c-1)$
  $c$ : number of checks per block
  $r$ : number of blocks = minimum number of replicates for a check
- The minimum number of blocks should be $\geq \left(\frac{10}{c-1}\right) + 1$
- example, using 4 checks :
  $\left(\frac{10}{4-1}\right) + 1 = 4.33 \approx 5$ blocks
- Each block contains at least $c+1$ plots (ie at least one new accession assessed per block)

# Analysis

- the global experimental error is estimated in the checks' analysis in a RCBD
- MSE , $\sigma_e^2$ will be used to build standard errors for the comparisons of the means
- Phenotypic values of 'new' entries will be computed
  - by 'adjusting means for block effect' if fixed models are used (not seen in this course)
  - with BLUP if mixed models are used

ANOVA table for checks :

| Variation | ddl | SS | MS |
|---|---|---|---|
| block | r-1 | | |
| checks | c-1 | | |
| Résiduals | (r-1)(c-1) | | $\sigma_e^2$ |

# Standard Errors for comparisons of Means

- difference between two checks
  $S_c = \sqrt{2\sigma_e^2/r}$
- difference between adjusted means of two new entries in the same block
  $S_d = \sqrt{2\sigma_e^2}$
- difference between adjusted means of two new entries in two different blocks
  $S_v = \sqrt{2(c+1)\sigma_e^2/c}$
- difference bewteen adjusted means of a new entries and a check
  $S_{vc} = \sqrt{(r+1)(c+1)\sigma_e^2/rc}$

$c$=number of different checks per block ; $r$ = number of blocks = number of replicates of a check
We will use a Student '$t$' distribution with $(r-1)(c-1)$ ddl to compute CI.

# Example 1

- 8 new varieties are assessed using $c = 4$ cultivars as reference (checks)
- The design has 3 blocks (NOT optimal)
- $r = 3$ ; $c = 4$ thus the Student $t$ distribtution to use to compare means has $(r-1)(c-1) = 6$ ddl

- R practice

# Example 2

We intend to assess 30 new entries, using $c = 3$ genotypes as references (checks)

- number of blocks : $\frac{10}{c-1} + 1 = \frac{10}{2} + 1 = 6$
- number of new accessions per block : $30/6 = 5$
- total number of plots (over the 6 blocks) : $(5 + 3)x6 = 48$

▶ R practice – *by your own*

# Other 'Augmented Designs'

- **augmented complete block** (Federer, 1956, 1961),
- augmented incomplete complete block (Federer, 1956, 1961),
- augmented row–column (Federer and Raghavarao, 1975 ; Federer et al., 1975),
- **augmented split plot** (Federer, 2005b),
- augmented split block (Federer, 2005a)

# Do NOT always compute means 'adjusted for block effect' !

**Design** : *i* accessions are evaluated in a Randomised **Complete** Block Design with 3 blocks.
Compute the mean of entry *i* over the 3 blocks, using the parameters of Anova :

$$\begin{aligned}
\text{Mean}_i &= \frac{(\hat{\mu} + \alpha_i + \beta_1) + (\hat{\mu} + \alpha_i + \beta_2) + (\hat{\mu} + \alpha_i + \beta_3)}{3} \\
&= \frac{3\hat{\mu} + 3\alpha_i + \beta_1 + \beta_2 + \beta_3}{3} \\
&= \hat{\mu} + \alpha_i + \frac{1}{3}(\beta_1 + \beta_2 + \beta_3) \\
&= \hat{\mu} + \alpha_i
\end{aligned}$$

because, by construction of the ANOVA model, we have $\sum \beta_j = 0$

- We got a "**true**" estimate of the mean value using *balanced designs*